

MA653P - Computational Financial Modelling Lab

Assignment 4

Instructions.

- This assignment builds upon the dataset collected in Assignment 1 and processed in Assignment 2. Students must **ONLY** use the data of N stocks gathered earlier (restrict $N = 20$ if it is more than 20) for their selected index.
- Use initial 70% part of the data as a seen data for model building (finding VAR, CVAR, portfolio construction etc.) and remaining 30% data for testing the performance.
- Submit a single `.ipynb` Jupyter notebook named `RollNo_MA653P_A04.ipynb`.
- Every cell must be executed top-to-bottom with no errors before submission. Include all output (tables, plots, printed values) in the saved notebook.
- Use **only** the following libraries: `numpy`, `scipy`, `pandas`, `matplotlib`, `seaborn`, `yfinance`, `cvxpy`. All are installable via `pip`.
- Each task must begin with a **Markdown cell** stating the task number, title, and a brief description of your approach.
- Figures must have titles, axis labels, and legends. Tables must have column headers.
- The evaluation will be based on **correctness**, **clarity of code**, **quality of output**, **written observations**, and **responses to the questions asked**.
- The goal is not just code that runs, but code that *explains* — through outputs, plots, and commentary.

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1. Using the analytical formulas derived in the theory assignment: $A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$, $B = \mathbf{1}^\top \Sigma^{-1} \boldsymbol{\mu}$, $C = \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu}$, $\Delta = AC - B^2$.
 - (a) **Closed-form GMV.** Implement the closed-form solution $\mathbf{w}_{\text{GMV}} = \Sigma^{-1} \mathbf{1} / A$ and compute $\mu_{\text{GMV}} = B/A$, $\sigma_{\text{GMV}} = 1/\sqrt{A}$.
 - (b) **Numerical QP verification.** Solve the same problem using `scipy.optimize.minimize` (method `SLSQP`) with the budget constraint $\mathbf{1}^\top \mathbf{w} = 1$ and long-only $w_i \geq 0$. Verify that the numerical weights agree with the analytical solution to four decimal places.
 - (c) **Efficient frontier.** For 200 target returns $m \in [\mu_{\text{GMV}}, 1.5 \max_i \mu_i]$, solve the constrained QP and collect $(\sigma(m), m)$. Plot the efficient frontier in the (σ, μ) plane. Mark the GMV portfolio with a red star.
 - (d) **Weight bar chart.** Plot a horizontal bar chart of the GMV weights. Which sector dominates the GMV portfolio? Explain this result using the theory of diversification.
 2. Use the training-period equal-weight portfolio $\mathbf{w}_0 = \mathbf{1}/N$ as the reference portfolio.
 - (a) **Historical simulation.** Compute $\widehat{\text{VaR}}_{99\%}$ and $\widehat{\text{CVaR}}_{99\%}$ by sorting the T daily portfolio losses and applying:

$$\widehat{\text{VaR}}_\alpha = L_{(\lceil \alpha T \rceil)}, \quad \widehat{\text{CVaR}}_\alpha = \frac{1}{n_{\text{tail}}} \sum_{i=1}^{n_{\text{tail}}} L_{(i)}, \quad n_{\text{tail}} = \lceil (1 - \alpha)T \rceil.$$

Report daily and annualised values in rupees for a portfolio worth Rs. 1,00,00,000.

- (b) **Parametric Normal.** Estimate $(\hat{\mu}_P, \hat{\sigma}_P)$ from the equal-weight portfolio and compute $\text{VaR}_\alpha = -\hat{\mu}_P + z_\alpha \hat{\sigma}_P$, $\text{CVaR}_\alpha = -\hat{\mu}_P + \hat{\sigma}_P \phi(z_\alpha)/(1 - \alpha)$.
- (c) **Parametric Student- t .** Fit a Student- t_ν distribution to portfolio losses using `scipy.stats.t.fit`. Report the fitted (ν, μ, σ) and compute $\text{VaR}_{99\%}$ and $\text{CVaR}_{99\%}$ under this distribution.
- (d) **Comparison table and chart.** Produce a 3×4 table (rows: Historical / Normal / Student- t ; columns: $\text{VaR}_{95\%}$, $\text{VaR}_{99\%}$, $\text{CVaR}_{95\%}$, $\text{CVaR}_{99\%}$). Plot the loss histogram with KDE and mark all six VaR/CVaR thresholds.

3. Implement all three models below using the training data. Use $\alpha = 99\%$, $\tau = r_f/252$ (daily), $r_f = 6\%$, long-only $w_i \geq 0$, $\mathbf{1}^\top \mathbf{w} = 1$, and target return $m = 12\%$ per annum.

Model	Objective	Auxiliary vars	Solver
Minimum Variance	$\min \mathbf{w}^\top \Sigma \mathbf{w}$	None	SLSQP
CVaR Minimisation	$\min \zeta + \frac{1}{(1-\alpha)T} \sum_t u_t$	$\zeta, \mathbf{u}$	linprog
Semivariance	$\min \frac{1}{T} \sum_t d_t^2$	$\mathbf{d}$	SLSQP

- (a) Implement each model as a separate Python function that accepts $(R_{\text{train}}, \boldsymbol{\mu}, \Sigma, r_f, m, \alpha, \tau)$ and returns the optimal weight vector $\mathbf{w}^*$.
- (b) For each model report: $\mathbf{w}^*$ (bar chart), μ_P , σ_P , Sharpe Ratio (SR), $\text{VaR}_{99\%}$, $\text{CVaR}_{99\%}$ on the *training set*.
- (c) Plot all four weight vectors side-by-side as a grouped bar chart with stock tickers on the x-axis.
- (d) In a Markdown cell, explain why Model 2 (CVaR) typically concentrates less than M1 (Variance) in the tail of the loss distribution.
4. Repeat Task 3 adding the cardinality constraint $K = \lceil N/3 \rceil$ (at most K non-zero weights), where N is the number of stocks.

- (a) **Implement cardinality via branch-and-bound.** Write a function `card_solve(model_fn, R, K, ...)` that enumerates all $\binom{N}{K}$ subsets of size K , solves the unconstrained model on each subset, and returns the globally optimal sparse weight vector.

For $N = 20$, $K = 7$: $\binom{20}{7} = 77,520$ subsets. This is feasible in <5 minutes.

- (b) For each of the four models, solve the cardinality-constrained version and report the selected K stocks and their weights.
- (c) Produce a summary table comparing unconstrained vs. cardinality results:

Variant	μ_P	σ_P	SR	$\text{CVaR}_{99\%}$
Model 1 (unconstrained)				
Model 1 (budget + cardinality)				
Model 2 (unconstrained)				
Model 2 (budget + cardinality)				
Model 3 (unconstrained)				
Model 3 (budget + cardinality)				

- (d) Comment: does cardinality always increase risk? Under what conditions can it *reduce* CVaR?
5. The following must be assessed on depth of analysis, not length.
- (a) **Data observations.** Comment on the return distribution of the stocks data: fat tails, skewness, and sector-level patterns.
- (b) **Model comparison.** Based on the metrics table in Task 4, rank the four models on each metric.
- (c) **Effect of cardinality.** How does adding $K = \lceil N/3 \rceil$ affect performance?
- (d) **Model recommendation.** Which single model (with or without cardinality) would you recommend to a risk manager at an equity fund? Justify based on:
- theoretical properties (coherence, convexity, robustness),
 - practical considerations (computation time, turnover, number of holdings),
 - empirical performance on the data.

Submission checklist.

- ☐ Notebook runs end-to-end without errors
- ☐ All the above tasks have a Markdown header cell
- ☐ File named RollNo_MA653P_A04.ipynb